ZetaERP — Manual Bug-Testing SOP

Part 4 — Sales (Quotations, Orders, Invoices, Credit Notes, Challans)

Document	QA-SOP-04 — Sales
Product	ZetaERP / Zenith (multi-tenant POS / ERP)
Environment under test	Production — https://erp.zetaclantech.com
Scope of this part	The **Sales** module: Dashboard, Quotations, Sales Orders, **Tax Invoices** (GST), Credit/Debit Notes, Delivery Challans, Payments, Commissions, Warranty, Sales Forecast, and the print/PDF outputs. (POS counter billing is Part 3; Accounting ledgers are separate.)
Audience	Manual QA tester (no code knowledge assumed)
Document owner	Platform / Solo-dev
Status	Active
Last updated	2026-07-01

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1. Purpose

This Standard Operating Procedure tells a tester **exactly how to manually test the Sales module end-to-end (A→Z)** — the full **order-to-cash** flow (Quotation → Order → Delivery Challan → Invoice → Credit Note) plus payments, commissions, warranty and forecasting — including the

edge cases, the **GST-correctness** cases, the **multi-tenant isolation** cases, the **security** cases, and the **visual/UI** cases, and **how to record a defect** when behaviour differs from the *Expected Result*.

Sales documents are **legal and financial records** (tax invoices, credit notes). Test deliberately, and never on live data.

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2. Critical rules for the tester (read before starting)

> ⚠️ **This is a LIVE commercial system producing legal GST documents.**

1. **NEVER create, edit, renumber, or delete a real invoice / credit note** on the live flagship tenant **MARUTHI (org `100000`)** or any live tenant. An issued tax invoice is a legal record. Do all create/edit/convert testing on a **throwaway tenant** you control.
2. ⚠️ **Known gap — issued invoices are editable.** There is currently **no lock** preventing edits/deletes of line items on a finalized/paid invoice (see §17). This makes rule #1 doubly important: do not test the "edit issued invoice" cases on real data.
3. **Never fabricate** legal/tax data — GSTIN, HSN/SAC codes, PAN, e-invoice IRN, e-way bill numbers, real customer details. Use clearly-fake QA values on a throwaway tenant.
4. **Multi-tenant isolation is a key surface** (§14). A tenant must never see or edit another tenant's quotes/orders/invoices. Any cross-tenant access is **Critical**.
5. **Clean up every test document** (quote, order, invoice, credit note, challan, commission, warranty) you create — or hand the list to the dev. Issued invoices may not be deletable; prefer cancelling/voiding test ones.
6. **Don't email/WhatsApp/print-and-send** a test invoice to a real address. PDFs may be generated; keep them local.

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3. Test environment & preconditions

Open from the app menu → **Sales**. Screen URLs (`/apps/zenith/sales/…`):

Area	Screens
Dashboard	`SalesDashboard`
Quotations	`FindQuotation` , `CreateQuotation` , `EditQuotation?orderId=`
Orders	`FindOrder` , `CreateOrder` , `EditOrder?orderId=`
Invoices	`FindInvoice` , `CreateInvoice` , `EditInvoice?invoiceId=`
Credit/Debit Notes	`FindCreditNote` , `CreateCreditNote` , `EditCreditNote?invoiceId=`
Delivery Challans	`FindDeliveryChallan` , `CreateDeliveryChallan` , `EditDeliveryChallan?…`
Other	`SalesForecast` , `Commissions` , `WarrantyPolicies`
Print (PDF)	`InvoicePrint` , `OrderPrint` , `CreditNotePrint` , `ChallanPrint`

Log in as a tenant user so an **active organization** scopes the lists. For isolation tests you need **two tenants**. Have at least **two customers in different states** (for intra- vs inter-state GST) and a couple of **products** set up.

Browsers: full pass on Desktop Chrome; spot-check Edge/Safari/mobile. **Tools:** DevTools (Console/Network), a calculator (to check GST math), a QA inbox.

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4. Test data

Label	Use
Throwaway tenant	Non-MARUTHI org for all create/edit
`CUST_SAME_STATE`	Customer in the **same** state as the seller (→ CGST+SGST)
`CUST_OTHER_STATE`	Customer in a **different** state (→ IGST)
`PROD_A` / `PROD_B`	Two products with known prices + HSN + GST rate

Label	Use
Fake GSTIN	`29ABCDE1234F1Z5`-style dummy (throwaway only)
Amounts to verify GST	e.g. qty 10 × ₹100 @ 18% → taxable 1000, tax 180

5. Conventions & severity

Case Type: `FUNC` · `EDGE` · `GST` (tax-correctness) · `UI` · `A11Y` · `SEC` · `ISO` · `PERF`.

Sev	Meaning	Example
S1 — Critical	Wrong tax math on a legal doc, cross-tenant access, altering/duplicating an issued invoice number, data loss	IGST charged on an intra-state sale; tenant A opens tenant B's invoice
S2 — High	Major flow broken	Convert-to-invoice fails; totals don't add up; can't save an invoice
S3 — Medium	Wrong behaviour, limited impact	Negative qty accepted; no GSTIN format check; status not enforced
S4 — Low	Cosmetic / minor UX	Column misaligned; PDF spacing off

Logging a defect (template §16): Case ID, steps, expected vs actual (show the **GST math** for tax cases), tenant, browser, screenshot, Console/Network capture.

6. Flow map (order-to-cash)

```
QUOTATION ──convert──► ORDER ──convert──► INVOICE ──► (payment)
(OrderProposed)        (OrderPlaced)      (InvoiceInProcess→…→Paid)
     |                     |                  |
     |                     └─ OrderItemBilling links items   └─► CREDIT/DEBIT NOTE (from invoice)
     |
DELIVERY CHALLAN ──convert──► INVOICE   (goods movement w/o sale: job work, approval, transfer)

GST per line:  same state  → CGST + SGST  (½ rate each)
               other state → IGST          (full rate)
Numbering:     invoice no. = per-org sequence; gstInvoiceNumber auto "INV/<FY>/<id>" if blank (editable)
```

7. SALES DASHBOARD

Screen: `SalesDashboard`. Total Sales, Active Quotes, Total Customers, Generated Invoices, a 6-month trend, paid-vs-credit pie, and a 7-day forecast.

ID	Type	Title	Steps	Expected result	Sev
SL-001	FUNC	Dashboard renders	Open the dashboard	All cards + charts render; numbers present	S3
SL-002	FUNC	Figures are org-scoped & correct	Cross-check Total Sales / Generated Invoices against the invoice list	Match **your** org only; cancelled invoices excluded from sales	S2
SL-003	FUNC	Quick links	Click "recent order" / nav buttons	Navigate to the right screen/document	S3
SL-004	UI	Empty state	New tenant with no sales	Cards show 0; charts don't break; forecast shows a "need ~2 weeks of history" message	S3

8. QUOTATIONS & ORDERS

A **Quotation** is an order in status `OrderProposed`; an **Order** is `OrderPlaced`. Both add line items with automatic GST.

8.1 Quotations

ID	Type	Title	Steps	Expected result	Sev
SL-010	FUNC	Create quotation	CreateQuotation → pick customer, date, terms → Save	Quote created (OrderProposed); opens EditQuotation	S2
SL-011	FUNC	Customer list org-scoped	Open the customer dropdown	Only **your** org's customers	S1
SL-012	FUNC	Add line item + auto price	EditQuotation → add a product, leave unit price blank	Price auto-fills from the price list; line total = qty × price	S2
SL-013	GST	Intra-state split	Customer in the **same** state, GST 18%	Line shows **CGST 9% + SGST 9%**, no IGST	S1
SL-014	GST	Inter-state split	Customer in a **different** state, GST 18%	Line shows **IGST 18%**, no CGST/SGST	S1
SL-015	GST	Rate options	Set GST rate via dropdown	Only 0/5/12/18/28% selectable; tax recomputes	S3
SL-016	FUNC	Convert to order	Quote → "Convert to Order"	Status → OrderPlaced; opens EditOrder; items carry over	S2
SL-017	FUNC	Find quotations	FindQuotation, search by no./customer/status	Lists only OrderProposed/OrderRejected for your org	S2
SL-018	EDGE	Quote with no items	Create and try to convert an empty quote	Handled gracefully (no crash); ideally warns	S3

8.2 Orders

ID	Type	Title	Steps	Expected result	Sev
SL-020	FUNC	Create order	CreateOrder → customer, date, terms (RCM/Export/SEZ flags) → Save	Order created (OrderPlaced)	S2
SL-021	FUNC	Add items	EditOrder → add products	Items added with price	S2
SL-022	FUNC	Convert to invoice	EditOrder → "Convert to Invoice"	A new invoice is created from the order's product items; GST **recomputed**; lands on EditInvoice	S1
SL-023	FUNC	Double-invoice guard	Convert the **same** order to invoice twice	Second attempt blocked: "This order has already been invoiced (…)."	S2
SL-024	EDGE	Convert order with no products	An order with no billable product items → Convert	Blocked: "Order has no billable product items."	S3
SL-025	FUNC	Find orders	FindOrder, search/filter	Lists orders (excludes proposed/rejected/cancelled), org-scoped	S2
SL-026	ISO	Cross-tenant order	As tenant A, open `EditOrder?orderId=<tenant-B order>`	**Blocked** — "Order not found for your account."	S1

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9. TAX INVOICES (GST) — the core

Screens: `CreateInvoice`, `EditInvoice?invoiceId=`, `FindInvoice`. The invoice carries the legal GST data, line items, e-invoice/transport details, and status.

9.1 Create & line items

ID	Type	Title	Steps	Expected result	Sev
SL-030	FUNC	Create invoice	CreateInvoice → buyer, invoice type (Tax/BillOfSupply/Export), date → Save	Invoice created (InvoiceInProcess); opens EditInvoice	S2
SL-031	FUNC	Auto invoice number	Leave GST Invoice Number blank	Auto-generated `INV/<FY>/<id>` (e.g. `INV/26-27/00001234`); FY matches the company's financial year	S2
SL-032	FUNC	Add line item	EditInvoice → product, description, HSN/SAC, qty, unit price, GST rate → Add Item	Item added; taxable = qty×price (− discount); GST split correct	S2
SL-033	GST	Header totals aggregate	Add 2–3 items	Header **Total Taxable, Total CGST, Total SGST, Total IGST** equal the sum of the lines	S1
SL-034	GST	Intra vs inter-state	One invoice to a same-state buyer, one to other-state	CGST+SGST vs IGST exactly as in SL-013/014	S1
SL-035	FUNC	Delete line item	Delete an item	Item removed; header totals + GST **recompute**	S2
SL-036	FUNC	Round-off	Add items that produce a fractional grand total	A round-off line appears; payable is the rounded figure; amount-in-words matches	S2
SL-037	FUNC	Save tax details	Set Buyer/Supplier GSTIN, Place of Supply, RCM, Export, LUT/Bond, Shipping Bill → Save	Persist	S2
SL-038	FUNC	Save e-invoice/transport	Set IRN, E-Way Bill, Transporter, Vehicle, LR No. → Save	Persist	S3

9.2 GST-correctness (do the math by hand)

ID	Type	Title	Data	Expected result	Sev
SL-040	GST	18% intra-state	qty 10 × ₹100 @18%, same state	taxable 1000; CGST 90; SGST 90; total tax 180; grand 1180	S1
SL-041	GST	18% inter-state	same line, other state	taxable 1000; IGST 180; grand 1180	S1
SL-042	GST	Line discount	qty 10 × ₹100, 10% discount, 18%	taxable 900; tax 162; grand 1062	S1
SL-043	GST	Mixed rates	items at 5%, 12%, 18%	each line taxed at its own rate; header sums correct	S1
SL-044	GST	0% / exempt	a 0% line	no tax added; appears on the invoice correctly	S3
SL-045	GST	Missing buyer state	buyer has no tax registration / region	**PARTLY FIXED:** intra/inter-state now uses the parties' **registered state** (tax-registration `regionGeoId`), not a broken address lookup (which used to crash or always pick IGST). If a party has **no tax registration**, region is unknown → still defaults to **IGST** — set up GSTINs so the split is correct	S2
SL-046	EDGE	Negative quantity	qty = −5	**Record actual** — there is **no negative-qty check**; the line and totals go negative. Flag (S3)	S3
SL-047	EDGE	Zero qty / zero price	qty 0 or price 0	**Record actual** — no zero-amount check; an empty/zero invoice can be created	S3
SL-048	EDGE	Huge / decimal qty	qty 999999 / 2.5	Handled; precision = 2 decimals, HALF_UP	S4

9.3 Status, numbering integrity, find

ID	Type	Title	Steps	Expected result	Sev
SL-050	FUNC	Status change	Set status InProcess → Approved → Sent → Paid	Status updates; reflected in the list	S3

ID	Type	Title	Steps	Expected result	Sev
SL-051	EDGE	Edit a finalized/issued invoice	Set status past In-Process (Finalized/Sent/etc.), then try to add/delete a line item	**FIXED:** line items can only change while the invoice is a **draft (In-Process)**. Once issued, add/delete is blocked ("This invoice has been issued and its line items can no longer be changed. Raise a credit or debit note to correct it."). Drafts remain fully editable	S1
SL-052	EDGE	Manual duplicate number	Set `gstInvoiceNumber` to a value already used	**Record actual** — no uniqueness check is visible; a **duplicate invoice number** is a legal problem (S1)	S1
SL-053	FUNC	Find invoices	FindInvoice, search by no./customer/GSTIN/status	Org-scoped list; columns incl. IRN; ledger link	S2
SL-054	ISO	Cross-tenant invoice	As tenant A, open `EditInvoice?invoiceId=<tenant-B>`	**Blocked** — "Invoice not found for your account." (verify the add/delete/update actions are all guarded)	S1

10. CREDIT / DEBIT NOTES

Screens: `CreateCreditNote`, `EditCreditNote`, `FindCreditNote`. A credit note reduces tax liability (sales return etc.); a debit note increases it.

ID	Type	Title	Steps	Expected result	Sev
SL-060	FUNC	Create credit note	CreateCreditNote → pick original invoice, reason, date → Save	CN created from the original invoice (org/customer carried over)	S2
SL-061	FUNC	Reason captured	Pick a reason (Sales Return / Post-sale Discount / etc.)	Stored and shown	S3
SL-062	FUNC	Debit note	Same, choose "Debit Note"	Created as a debit note (increases liability)	S2
SL-063	GST	CN GST split	Add adjustment items	Same intra/inter-state GST rules as invoices	S1
SL-064	ISO	Original invoice must be yours	Try to create a CN against another tenant's invoice id	Blocked: "Original Invoice not found."	S1
SL-065	EDGE	Self / circular reference	Reference a credit note as its own original	**Record actual** — no circular-reference check; flag if it allows it	S3
SL-066	FUNC	Find credit notes	FindCreditNote, filter by type/customer/original invoice	Org-scoped; shows Credit vs Debit, original invoice id	S3

11. DELIVERY CHALLANS

Screens: `CreateDeliveryChallan`, `EditDeliveryChallan`, `FindDeliveryChallan`. Goods movement without a sale (job work, approval, transfer), with e-way bill.

ID	Type	Title	Steps	Expected result	Sev
SL-070	FUNC	Create challan	CreateDeliveryChallan → receiver, type, date → Save	Challan created (Draft); auto-number `CH-<id>` if blank	S2
SL-071	FUNC	Add goods	EditDeliveryChallan → product, HSN, qty, UOM, est. value → Add	Items added	S2
SL-072	FUNC	Transport / e-way bill	Set E-Way Bill, Vehicle, Transporter, LR → Save	Persist	S3
SL-073	FUNC	Convert to invoice	Convert challan → invoice	Invoice created from challan items; challan → Converted	S2
SL-074	FUNC	Status flow	Draft → Dispatched → Delivered / Returned / Cancelled	Status updates	S3
SL-075	ISO	Find challans org-scoped	FindDeliveryChallan	Only your org's challans	S1

12. PAYMENTS · COMMISSIONS · WARRANTY · FORECAST

ID	Type	Title	Steps	Expected result	Sev
SL-080	FUNC	Payment link	Generate a payment link for an invoice (Razorpay/CCAvenue)	A link is created (currently a **mock** gateway — no real charge); status CREATED	S3
SL-081	FUNC	UPI QR	Generate a UPI QR for an invoice	A valid `upi://pay?...` string with the right payee/amount	S3
SL-082	EDGE	⚠ Payment doesn't update invoice	Simulate a "paid" callback	**Record actual — KNOWN GAP:** the payment-link status flips to PAID but the **invoice status does not become Paid** and no Payment is recorded against it. Flag (S2)	S2
SL-083	FUNC	Commission structure	Commissions → define a salesperson rate (%) and validity	Structure created (org-scoped)	S3
SL-084	FUNC	Commission calc	Calculate commission for an order with a salesperson	commission = order grand total × rate ÷ 100	S3
SL-085	FUNC	Warranty policy + register	WarrantyPolicies → create a policy, register a warranty (serial, purchase date)	Warranty registered; end date = purchase + duration	S3
SL-086	FUNC	Warranty claim	File a claim on an **active** warranty	Claim created	S3
SL-087	EDGE	Claim on expired warranty	File a claim after the warranty period	Should reject ("Warranty is not active"); **verify** it actually checks expiry, not just status	S3
SL-088	FUNC	Sales forecast	SalesForecast → run	Produces a 7/14-day projection with a confidence %; with <~2 weeks of data shows the "need more history" message	S3
SL-089	ISO	Forecast/commission org-scoped	View as two tenants	Each sees only its own	S1

13. PRINT / PDF

Screens: `InvoicePrint`, `OrderPrint`, `CreditNotePrint`, `ChallanPrint` (PDF outputs).

ID	Type	Title	Steps	Expected result	Sev
SL-090	FUNC	Invoice PDF	Print an invoice	PDF shows seller/buyer, GSTIN, line items, **GST split**, totals, **amount in words** (Indian lakh/crore), invoice no. + date	S2
SL-091	FUNC	Other PDFs	Print a quotation/order, credit note, challan	Each renders correctly with the right title	S3
SL-092	ISO	Only buyer/supplier may print	As an unrelated tenant, hit `InvoicePrint?invoiceId=<other>`	Denied / blank — only the supplier or buyer org can print	S1
SL-093	UI	PDF layout	Inspect a multi-page invoice	No clipped columns; totals on the right page; numbers aligned	S4

14. MULTI-TENANT ISOLATION (high priority)

ID	Type	Title	Steps	Expected result	Sev
ISO-001	ISO	Quotations/Orders scoped	Tenant A lists	Only A's quotes/orders (`vendorPartyId`)	S1
ISO-002	ISO	Invoices scoped	Tenant A lists invoices	Only A's invoices (`fromPartyId`)	S1
ISO-003	ISO	Direct-URL invoice/order/CN	A opens B's `EditInvoice` / `EditOrder` / `EditCreditNote` by id	Blocked ("not found for your account")	S1

ID	Type	Title	Steps	Expected result	Sev
ISO-004	ISO	Add/delete item cross-tenant	(Dev-assisted) POST addItem/deleteItem for B's invoice as A	Blocked by the same ownership guard	S1
ISO-005	ISO	Challans/Commissions/Warranty/Forecast	Each scoped to the active org	No cross-tenant rows	S1
ISO-006	ISO	Print cross-tenant	A prints B's document	Denied	S1
ISO-007	ISO	Customer/product pickers scoped	Open the dropdowns in any create screen	Only your org's customers/products	S1

15. UI / ACCESSIBILITY / SECURITY

ID	Type	Title	Check	Expected result	Sev
SL-100	UI	Desktop layout	All screens at 1920/1366	Item grids and totals aligned; no overflow	S4
SL-101	UI	Mobile layout	Create/edit at 360–390px	No horizontal scroll; item form usable; totals readable	S3
SL-102	A11Y	Keyboard	Tab through an add-item form	Logical order; Enter adds; visible focus	S3
SL-103	SEC	XSS in free text	Put `<img src=x onerror=alert(1)>` in description / customer name	Rendered/escaped as text, never executes (incl. on the PDF)	S1
SL-104	SEC	CSRF	(Dev-assisted) POST a create/addItem without the session token	Rejected	S2
SL-105	EDGE	Double-submit	Click "Add Item" / "Convert" twice fast	Only one item/invoice created	S2

16. Cross-flow / regression (full order-to-cash) & defect template

ID	Type	Title	Expected result	Sev
XF-001	FUNC	Quote → Order → Invoice	Create a quote, convert to order, convert to invoice; GST recomputes at each stage; totals consistent	S1
XF-002	GST	State change mid-flow	Change the customer's state between order and invoice → GST flips CGST/SGST ↔ IGST correctly	S1
XF-003	FUNC	Invoice → Credit Note	Raise a CN against the invoice; amounts and GST reverse correctly	S1
XF-004	FUNC	Challan → Invoice	Goods on a challan convert into a correct invoice	S2
XF-005	ISO	End-to-end isolation	Nothing tenant A creates is visible/editable by tenant B at any stage	S1

```
Defect ID:        ZE-SALES-____
Linked Case ID:   SL-0__ / ISO-0__ / XF-0__
Title / Severity / Type:
Environment:      Prod | <browser> | tenant: <org>
Steps / Expected / Actual:   (for GST cases, show the math: taxable, CGST/SGST/IGST, grand)
Document ids touched (quote/order/invoice/CN/challan):
Evidence:         screenshot + Console/Network (mandatory for GST, ISO, SEC)
```

17. Areas to probe hardest (attention map)

These are the most likely defects in this module — spend extra time here:

1. **GST correctness (§9.2, XF-002)** — the single most important thing. Intra vs inter-state, discounts, mixed rates, and the **"missing state → defaults to IGST"** behaviour (SL-045). Do the math by hand.
2. **Issued-invoice immutability (SL-051)** — *fixed*: line items are locked once the invoice leaves draft.
3. **GST intra/inter-state (SL-045)** — *fixed*: now uses the tax-registration state (`regionGeoId`), removing the crash and the always-IGST fallback. **Separate open bug found here:** the manual EditInvoice "Add Item" doesn't persist a line at all (`create#InvoiceItem` no-ops silently) — the Order→Invoice path works, but manual add on an invoice needs a fix.
4. ⚠ **Invoice-number uniqueness (SL-052)** — manual `gstInvoiceNumber` has no duplicate check.
5. ⚠ **Payment doesn't close the invoice (SL-082)** — payment link → PAID, but the invoice stays unpaid; no Payment recorded.
6. **Validation gaps** — negative qty (SL-046), zero amount (SL-047), no GSTIN/HSN format check, no credit-limit enforcement, RCM/Export flags stored but **not applied** to GST.
7. **Multi-tenant isolation (§14)** — lists and document actions are guarded (`fromPartyId` / `vendorPartyId`); confirm the *edit/add/delete/print* paths are **all** guarded, not just the lists.

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18. Teardown (mandatory)

1. **List every test document** (quote/order/invoice/CN/challan/commission/warranty) with ids.
2. **Cancel/void** test invoices and credit notes (don't leave fake tax documents around); hand the list to the dev for proper cleanup.
3. Delete any **PDFs** generated on the test machine.
4. Confirm you **never created/edited/numbered a document on MARUTHI** (org `100000`) or any live tenant.
5. Attach the run sheet(s) + defect ids to this part's report.

End of Part 4 — Sales. Related: Part 1 — Login & Onboarding; Part 2 — Contacts; Part 3 — Settings. Sales consumes customers/products from Contacts & Inventory and posts to Accounting; the GST-correctness and issued-invoice-immutability findings here should be triaged with the dev before any live use.